

Coding practice: case_when, tables

Your name

0. Change your name in the YAML and load in the packages necessary for making beautiful tables. Then run the following chunk.

```
library(tidyverse)
# add additional packages here

diamonds2 <- read_csv("https://raw.githubusercontent.com/midd-stat201a-fall25/midd-stat201a-fall25/main/diamonds2.csv")
diamonds_lab <- read_csv("https://raw.githubusercontent.com/midd-stat201a-fall25/midd-stat201a-fall25/main/diamonds_lab.csv")
```

We will work with a modified version of the `diamonds` dataset that I have loaded for you above, called `diamonds2`. This is just like the `diamonds` data set, but now includes a variable called `id` that uniquely identifies the diamond.

You also loaded in another dataset called `diamonds_lab` that includes, for certain diamonds as identified by `diamond_id`, the laboratory (GIA, IGI, AGS) that certified the diamond's characteristics and authenticity.

1. Modify (i.e. store over) your `diamonds2` dataframe to:
 - include the certifying lab for the diamond, only retaining cases that have been certified, and
 - include an additional variable called `price_ct` which is the price of the diamond per carat

You can do this in one big pipeline, or two separate pipelines!

2. Create a beautiful table (i.e. not data frame) that displays the proportion of diamonds whose price per carat exceeds 8000 USD for each color of diamond. Your table should only display the color of the diamonds and the proportion that are greater than 8000 USD.
3. To your `diamonds2` data frame, use `case_when()` to add a variable called `carat_cat` that takes the value:

- “< 1” if the diamond is less than 1 carat
- “1-3” if the diamond is between 1 and 3 carats, inclusive
- “> 3” if the diamond is more than 3 carats

Then using an appropriate plot type, visualize the distribution of price per carat for each of the categories you created. Make sure your plot has informative axis labels. What do you notice?

Answer: